

Mathematics A

Unit 1 • Chapter OL-T58 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

9 classified items • 36 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T58 • Expertise Q16+ • 9 items • 36 marks

Chapter	Items	Marks	Starts
01. OL-T58 Combined & Conditional Probability	9	36	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Combined & Conditional Probability

Unit 1 • Expertise • 36 marks • 9 item(s)

June 2025 | 4MA1-1H Q21 | 3 marks

Calculate an exact composition in three or more draws

June 2025 | 4MA1-2H Q20 | 6 marks

Calculate a direct given-that probability

June 2024 | 4MA1-1H Q16 | 4 marks

Use a complement in repeated no-replacement selection

November 2023 | 4MA1-1H Q20 | 4 marks

Find exactly one, same or different independent outcomes

June 2022 | 4MA1-2H Q16 | 8 marks

Calculate a direct given-that probability

January 2020 | 4MA1-2H Q18 | 2 marks

Use a histogram or table as a conditional sample space

November 2021 | 4MA1-1H Q16 | 1 marks

Calculate a direct given-that probability

November 2021 | 4MA1-2H Q16 | 3 marks

Compare with-replacement and without-replacement models

May-Nov 2020 | 2H Q20 | 5 marks

Recover an unknown population from no-replacement probability

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q21

3 marks

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

.....
(Total for Question 21 is 3 marks)

WORKING SPACE

4MA1-2H Q20

6 marks

20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots

12 grow carrots and potatoes and tomatoes

18 grow carrots and potatoes

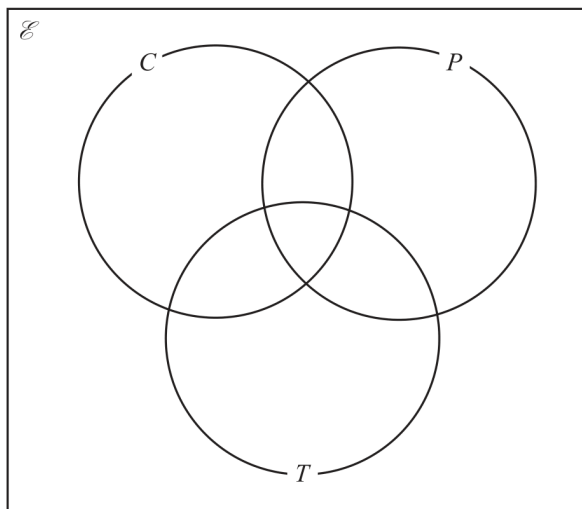
27 grow carrots and tomatoes

32 grow potatoes and tomatoes

29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

(2)

WORKING SPACE

4MA1-1H Q16

4 marks

16 There are 20 sweets in a box.

15 of the sweets are red

5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

WORKING SPACE

4MA1-1H Q20 demand

4 marks

20 A bag contains only 10 cent coins and 20 cent coins.

Josip takes at random a coin from the bag, records its value and replaces it in the bag. He then takes at random a second coin from the bag, records its value and replaces it in the bag.

Josip finds the mean value of the two coins.

The probability that the two coins have a mean value of 10 cents is $\frac{49}{121}$

Work out the probability that the two coins have a mean value of 15 cents.

.....

WORKING SPACE

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

8 marks

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

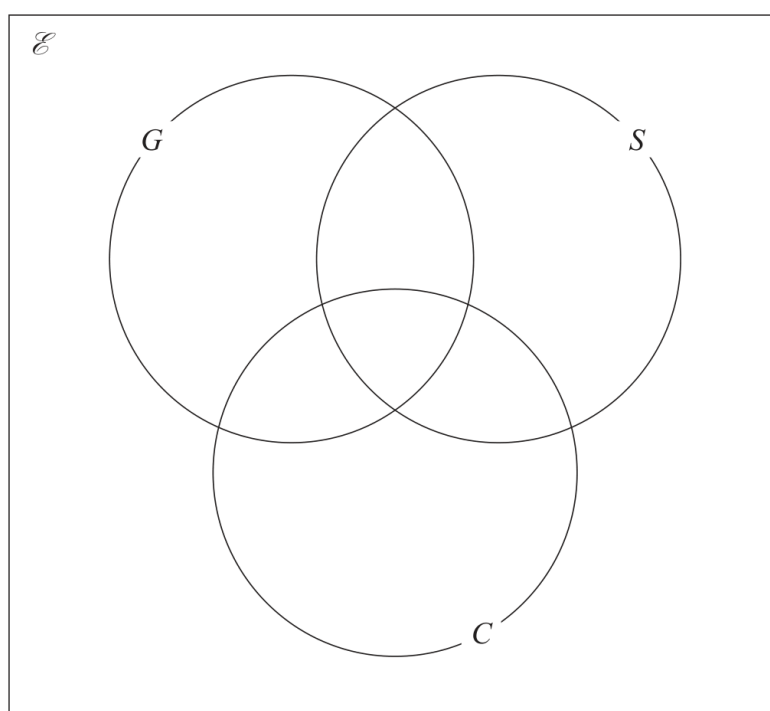
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

(b) Find

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

8 marks

(i) $n(G)$

.....
(1)

(ii) $n([G \cup S]')$

.....
(1)

(iii) $n(G' \cap C)$

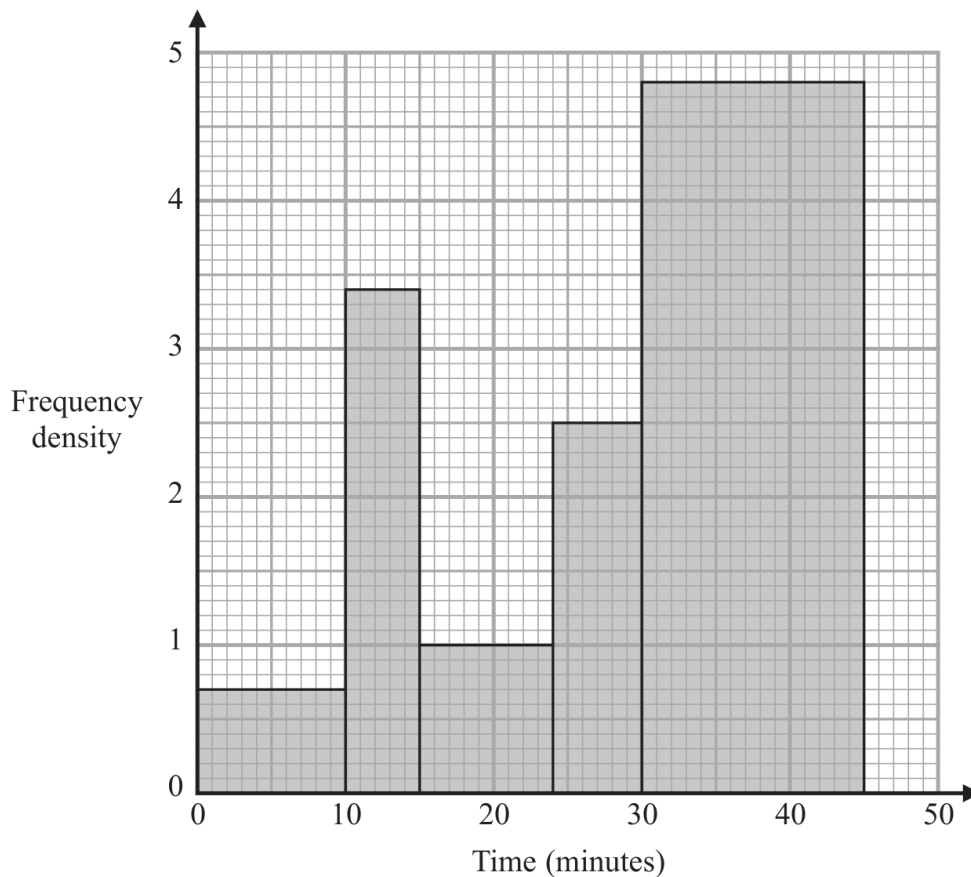
.....
(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

.....
(2)

- 18 The histogram gives information about the times, in minutes, that some customers spent in a supermarket.



One of the customers is selected at random.

Given that this customer had spent more than 30 minutes in the supermarket,

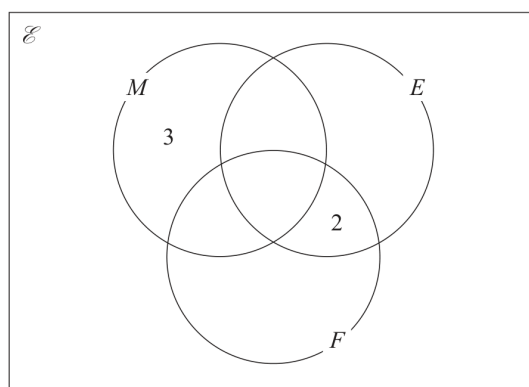
- (b) find the probability that this customer spent more than 36 minutes in the supermarket.

(2)

YOUR WORKING

- 16 There are 32 students in a class.
 In one term these 32 students each took a test in Maths (M), in English (E) and in French (F).
 25 students passed the test in Maths.
 20 students passed the test in English.
 14 students passed the test in French.
 18 students passed the tests in Maths and English.
 11 students passed the tests in Maths and French.
 4 students failed all three tests.
 x students passed all three tests.

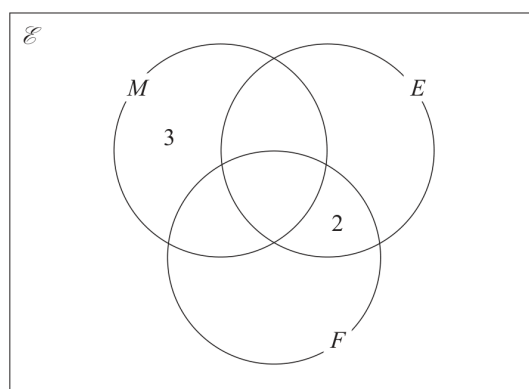
The incomplete Venn diagram gives some more information about the results of the 32 students.



- (a) Use all the given information about the results of students who passed the test in Maths to find the value of x .

$$x = \dots\dots\dots (2)$$

- (b) Use your value of x to complete the Venn diagram to show the number of students in each subset.



(2)

A student who passed the test in Maths is chosen at random.

- (c) Find the probability that this student failed the test in French.

$$\dots\dots\dots (1)$$

YOUR WORKING

16 A box contains 15 counters.

There are 4 red counters, 5 green counters and the rest are yellow counters.

Niklas takes at random a counter from the box and writes down the colour of his counter.
He then puts the counter back into the box.

Sasha then takes at random a counter from the box and writes down the colour of her counter.

Work out the probability that the counters taken by Niklas and Sasha both have the same colour.

YOUR WORKING

20 A bag contains X counters.

There are only red counters and blue counters in the bag.

There are 4 more blue counters than red counters in the bag.

Finty takes at random 2 counters from the bag.

The probability that Finty takes 2 blue counters from the bag is $\frac{3}{8}$

Work out the value of X .

Show clear algebraic working.

(Total for Question 20 is 5 marks)

YOUR WORKING
